

Program 6

Deploy an Automated Build Pipeline using Docker Hub

Program Structure:

```
program6/  
----->Dockerfile  
----->app.py  
----->requirements.txt
```

Procedure:

step 1: Create Dockerfile without any extensions
nano Dockerfile

GNU nano 6.2	Dockerfile
# Use Node base image	
FROM node:18	
# Set working directory inside container	
WORKDIR /app	
# Copy package files	
COPY package*.json ./	
# Install dependencies	
RUN npm install	
# Copy remaining files	
COPY . .	
# Expose port	
EXPOSE 3000	
# Start application	
CMD ["npm", "start"]	

step 2: create a simple requirements.txt file
Build an image
docker build -t program6

```

Activities Terminal Dec 31 10:19 1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx: ~/program6

bash: /home/pavithra/dev/flutter/bin/cache: No such file or directory
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~$ cd program6
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program6$ ls
Dockerfile index.js package.json
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program6$ docker build -t program6
ERROR: docker: 'docker buildx build' requires 1 argument
Usage: docker buildx build [OPTIONS] PATH | URL | -

Run 'docker buildx build --help' for more information
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program6$ docker build -t program6 .
[+] Building 3.3s (11/11) FINISHED docker:default
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 315B 0.0s
=> [internal] load metadata for docker.io/library/node:18 3.1s
=> [auth] library/node:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/5] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d39 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 92B 0.0s
=> CACHED [2/5] WORKDIR /app 0.0s
=> CACHED [3/5] COPY package*.json ./ 0.0s
=> CACHED [4/5] RUN npm install 0.0s
=> CACHED [5/5] COPY . . 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:6612e97d49eb8f5a77f1dc33b2c171bb7b1b785abee58 0.0s
=> => naming to docker.io/library/program-6 0.0s
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program6$ docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	SIZE	EXTRA
docker:latest	b21da390a210	132MB		0B	
gcr.io/k8s-minikube/ktcbase:v0.0.48	c6b5532e987b	1.31GB		0B	U
gcrh.io/pavithra282/docker:latest	b21da390a210	132MB		0B	
ghcr.io/pavithra282/docker:latest	b21da390a210	132MB		0B	
ghcr.io/pavithra282/hub:latest	5dfc227ce476	1.09GB		0B	
git-app:latest	392fe6e9de27	1.1GB		0B	U
hub:latest	5dfc227ce476	1.09GB		0B	
image1:latest	83a311f98efd	8.32MB		0B	U
image2:latest	105654a3038e	8.32MB		0B	U
jenkins/jenkins:lts-jdk11	2771c4579171	471MB		0B	U
mysql/mysql-server:5.6	8c587c89edee	238MB		0B	U
ninmls/apache-php5:latest	ebf63727818a	448MB		0B	U
node:latest	5cdc8ed4f59b	1.13GB		0B	
pavithradn/hub:latest	5dfc227ce476	1.09GB		0B	
pavithradn26/docker:latest	b21da390a210	132MB		0B	
pavithradn26/hub:latest	5dfc227ce476	1.09GB		0B	
pavithradn26/docker:latest	b21da390a210	132MB		0B	
prg3:latest	dc07be177ca9	133MB		0B	U
program-6:latest	6612e97d49eb	1.09GB		0B	U
program3:latest	dc07be177ca9	133MB		0B	U

Step 3: Run the container
 docker run -p 5000:5000 program6

```

program3:latest          dc07be177ca9          133MB          0B          U
program6:latest          6612e97d49eb          1.09GB         0B          U
quay.io/coreos/etcd:v3.5.13  bef2e3b99162          57.8MB         0B          U
registry:2               26b2eb03618e          25.4MB         0B          U
ubuntu:latest            c3a134f2ace4          78.1MB         0B
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program6$ docker run -p 3000:3000 program-6

> program-6@1.0.0 start
> node index.js

Server running on port 3000
^C
^C^C
got 3 SIGTERM/SIGINTs, forcefully exiting
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program6$ docker login
Authenticating with existing credentials... [Username: pavithradn26]

Info -> To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded

```

step 4: tag the image program6

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program6$ docker tag program-6 pavithradn26/program6:latest
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program6$ docker tag program6 pavithradn26/program6:latest
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program6$ docker push pavithradn26/program6:latest
The push refers to repository [docker.io/pavithradn26/program6]
ccc9a9817500: Pushed
4acee2e49d97: Pushed
37096354ae1a: Pushed
2ee51a854af5: Mounted from pavithradn26/hub
d2a991bcab4d: Mounted from pavithradn26/hub
b624aa2d5ea2: Mounted from pavithradn26/hub
d399c9dc306f: Mounted from pavithradn26/hub
84f9fa179c1b: Mounted from pavithradn26/hub
ce84ba212e49: Mounted from pavithradn26/hub
e4dc8cd9ecc8: Mounted from pavithradn26/hub
6428cc293366: Mounted from pavithradn26/hub
2f7436e79a0b: Mounted from pavithradn26/hub
latest: digest: sha256:6c80f94acd71eb6417837111394e6b5d5da656c28fe5336cb2ca34622bf9cd4f size: 2831
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program6$ git init
git add .
git commit -m "Initial commit"
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/1RV24MC073_pavithra/program6/.git/
[master (root-commit) ee64e99] Initial commit
```

step 5: push source code to Github

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program6$ git init
git add .
git commit -m "Initial commit"
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/1RV24MC073_pavithra/program6/.git/
[master (root-commit) ee64e99] Initial commit
3 files changed, 42 insertions(+)
create mode 100644 Dockerfile
create mode 100644 index.js
create mode 100644 package.json
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program6$ git remote add origin https://github.com/Pavithra282/program6.git
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program6$ git branch -M main
git push -u origin main
Username for 'https://github.com': Pavithra282
Password for 'https://Pavithra282@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 819 bytes | 819.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Pavithra282/program6.git
 * [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program6$ echo github_pat_11BJALSEY0b26ZERC1Je0R_yHTzC5jixS9gschn89ejQ8fMyyKHIEncwM6PBPQHOUiU3CYFHM
word-stdin

WARNING! Your credentials are stored unencrypted in '/home/1RV24MC073_pavithra/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/program6$
```

output

The screenshot shows a web browser displaying the GitHub repository page for 'program6' by user Pavithra282. The browser's address bar shows the URL 'github.com/Pavithra282/program6'. The repository page includes a navigation bar with links for Code, Issues, Pull requests, Actions, Projects, Wiki, Security, Insights, and Settings. The repository name 'program6' is displayed as 'Public'. Below the navigation bar, there are buttons for 'main' branch, '1 Branch', '0 Tags', a search bar 'Go to file', an 'Add file' button, and a 'Code' button. A commit history table shows the initial commit by Pavithra282, with files 'Dockerfile', 'index.js', and 'package.json' listed. At the bottom, there is a section titled 'Add a README' with a green button labeled 'Add a README'.

github.com/Pavithra282/program6

Pavithra282 / program6

<> Code Issues Pull requests Actions Projects Wiki Security Insights Settings

program6 Public Pin Watch 0

main 1 Branch 0 Tags Go to file Add file <> Code

Pavithra282 Initial commit ee64e99 - last week 1 Commit

Dockerfile	Initial commit	last week
index.js	Initial commit	last week
package.json	Initial commit	last week

README

Add a README

Help people interested in this repository understand your project.

Add a README